CD Drive Player

User Guide Rev. 1.2

Model: CDDP-100 Elehobica

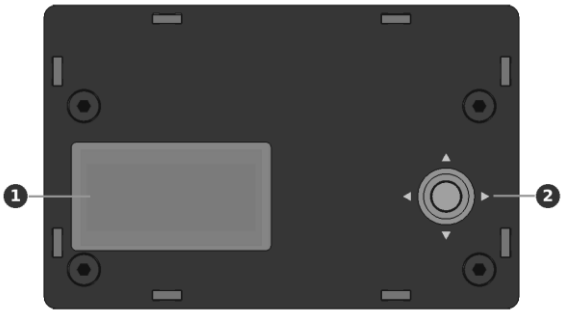
CD Drive Player turns a USB CD/DVD drive into an audio CD player. Connect a drive, insert an audio CD and listen through headphones, an amplifier or a digital audio device. No PC is needed.



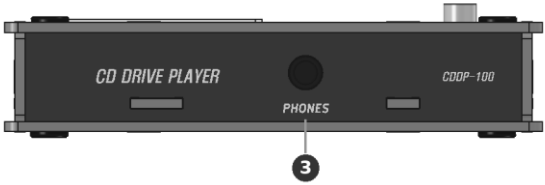
Features

- Plays audio CDs with a USB CD/DVD drive, without a PC
- Headphone and line output with volume control, or a fixed line output
- Optical digital output (S/PDIF, TOSLINK)
- OLED display and a 5-way switch for all operations
- Five play modes: Sequential, Shuffle, Sequential Repeat, Shuffle Repeat and Single Repeat
- No gap between consecutive tracks
- Resume from the stop position
- Powered from USB Type-C (POWER), and supplies USB bus power to the drive through the USB-A port (CD DRIVE USB)

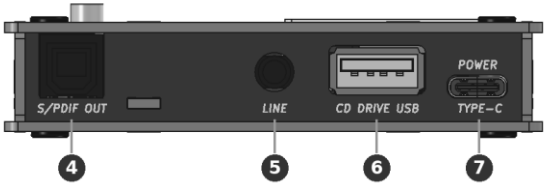
Panels



Top



Front



Rear

#	Name	Function
❶	**Display**	Shows the player state, the track and time, the volume and the play mode
❷	**5-way switch**	Tilt UP / DOWN / LEFT / RIGHT, or push the knob straight down (CENTER push). Two quick CENTER pushes are a CENTER double push
❸	**PHONES**	Headphone output, 3.5 mm stereo (with volume control)
❹	**S/PDIF OUT**	Optical digital output (TOSLINK)
❺	**LINE**	Line output to an amplifier, 3.5 mm stereo
❻	**CD DRIVE USB**	USB-A port for the USB CD/DVD drive
❼	**POWER**	USB Type-C power input, 5 V (3 A or more required, USB power adapter PD 20W or more recommended)

Specifications

Media	Audio CD (CD-DA)
Drive	USB CD/DVD drive, connected to CD DRIVE USB (USB-A) Any USB optical drive that reads CDs: CD-ROM, CD-RW, DVD-ROM, DVD±RW, Blu-ray combo drives, etc.
Analog outputs	PHONES: 3.5 mm stereo, volume-controlled (0 ~ 100) LINE: 3.5 mm stereo, volume-controlled (0 ~ 100) or fixed
Digital output	S/PDIF optical (TOSLINK), 16-bit / 44.1 kHz, no volume control
DAC	PCM5102A
Headphone Amp	TPA6132A2
Display	128 x 64 dot OLED
Controls	5-way switch (UP / DOWN / LEFT / RIGHT / CENTER push)
Power	USB Type-C, 5 V input. A power adapter rated 5 V / 3 A or more is required (USB power adapter PD 20W or more recommended)
Dimensions	83.0 (W) × 19.0 (H) × 53.0 (D) mm (excluding protrusions) 83.0 (W) × 21.0 (H) × 53.0 (D) mm (including the 5-way switch button)
Weight	78 g

Getting Started

What you need

Only the player is included (the CD/DVD drive bundle package also includes a CD/DVD drive). Please prepare the following:

- CD/DVD drive (not needed with the CD/DVD drive bundle package): any optical drive that reads CDs, such as a CD-ROM, CD-RW, DVD-ROM, DVD±RW or Blu-ray combo drive. This guide calls them "CD/DVD drive".
    - USB CD/DVD drive: connected directly. A slim (portable) drive is powered by the player.
    - SATA CD/DVD drive: can be used in some cases through a USB-SATA adapter. A desktop (5.25") drive needs its own 12 V power supply as well.
- USB Type-C power adapter, 5 V / 3 A or more. A USB power adapter PD 20W or more is recommended. A USB port of a PC is not recommended (most of them cannot supply 3 A).
    - A mobile battery (power bank) that supports USB PD is not recommended: it may not follow the current of the drive at the spin-up. Depending on the drive it may work; use it at your own judgment.
- Headphones, a 3.5 mm stereo cable to an amplifier, or an optical cable (TOSLINK) to a DAC or AV receiver

The power supply matters. Some drives need a large current while the disc spins up, and may stop even with a 5 V / 3 A adapter: the drive restarts and playback stops (see CHECK POWER in Troubleshooting). If the drive has an auxiliary power input (the power adapter of a USB enclosure or USB-SATA adapter, or the auxiliary power plug of a Y cable), it is recommended to power the drive from it directly.

Connection

1. If the drive has an auxiliary power input, connect it first: the auxiliary power plug of a Y cable (auxiliary power cable), or the power adapter of a USB enclosure or USB-SATA adapter (*).
2. Connect the USB cable of the drive to ❻ **CD DRIVE USB**, directly without a USB hub.
3. Connect the power adapter to ❼ **POWER**. The start screen appears, then **NO DISC** once the drive is recognized.
4. Connect headphones to ❸ **PHONES**, or connect ❺ **LINE** or ❹ **S/PDIF OUT** to your audio system. Connecting after the power-on avoids a pop noise.

(*) A 12 V supply is needed only for a desktop (5.25") drive. A slim drive runs from 5 V.



The start screen

Connect only the CD/DVD drive to CD DRIVE USB. Do not use a USB hub, and do not connect anything other than the CD/DVD drive.

Playing a disc

1. Open the tray: press the eject button of the drive, or **CENTER double push**.
2. Put an audio CD on the tray and close it. A drive with a motorized tray: **CENTER double push** (or the eject button of the drive) pulls the tray in. A slim drive: push the tray in by hand. **LOADING** is shown while the disc is read, then the number of tracks and the total time.
3. **CENTER push** to start playback.
4. **CENTER push** pauses playback (**PAUSE** blinks), and the next **CENTER push** continues it.
5. To stop, **CENTER push** and hold for 0.5 sec.
6. To take out the disc, **CENTER double push** or press the eject button of the drive.

Basic Operation

To do this	Operation
Start playback	CENTER push while stopped. If the resume is enabled, playback after a stop starts from the stop position (to change the resume: CONFIG > PLAY > RESUME)

To do this	Operation
Pause / Continue	CENTER push while playing / paused. If the resume is enabled, a pause longer than 600 sec ends in a stop
Stop	CENTER push and hold for 0.5 sec while playing / paused
Volume up / down	UP / DOWN (hold to repeat). 0 to 100
Track jump	LEFT / RIGHT. While playing: RIGHT goes to the next track, LEFT back to the start of the track (to the previous track within the first 1 sec of a track). While stopped: the first push shows track 1, further pushes select the track (wrapping around), and a CENTER push starts playback from it. The selection is cancelled after 10 sec without a button.
Fast Forward / Rewind	Hold RIGHT / LEFT while playing (faster after 4 sec)
Eject the disc	CENTER double push. It is recommended to stop playback first
Open / Close the tray (no disc)	CENTER double push (closing: a drive with a motorized tray only). With a motorized tray, a CENTER push with the tray open also closes it and starts playback

Display

The display has three parts. The example below shows a track being played.



Example: track 3 playing at 2:15, volume 30, play mode Sequential Repeat

The middle part shows the track number and the time. The top and bottom lines show the following:

Line	Display	Meaning
Top	**NO DRIVE**	No drive is recognized at CD DRIVE USB
	NO DISC	The drive is ready, but no disc is detected
	LOADING	The disc is being read. Takes a few seconds
	STOP	Stopped
	PLAY	Playing
	PAUSE (blinking)	Paused
	DISC ERROR	The drive cannot read this disc. Take it out
	CHECK POWER	Shown in turns with the state: the drive keeps restarting (see Troubleshooting)
Bottom	**VOL nnn**	PHONES and LINE outputs are volume-controlled

Line	Display	Meaning
	LINE	LINE output level is fixed. No PHONES output
	Icon (right)	The play mode (see Play mode icons)

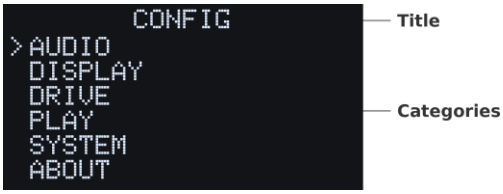
When the auto off is enabled, the display turns off when the player is stopped and no button is pressed for a while. Any button turns it on again; that push does nothing else. To change the auto off, use CONFIG > DISPLAY > AUTO OFF.

Play mode icons

Icon	Mode	Playback
	Sequential	Plays to the last track and stops (no icon)
	**Shuffle**	Plays every track once in a random order and stops
	**Sequential Repeat**	Repeats all tracks
	**Shuffle Repeat**	Repeats all tracks in a new random order every round
	**Single Repeat**	Repeats the current track

Configuration

CENTER push and hold for 2 sec while stopped to open the **CONFIG** menu. **UP** / **DOWN** move the cursor, **CENTER push** or **RIGHT** selects, **LEFT** goes back. A selected value applies at once and is saved. The menu closes after 5 sec without a button.



The CONFIG menu: the categories, with the cursor (>) on AUDIO

CONFIG > AUDIO

Item	Values	Description
ANALOG OUT	LINE + PHONES (VOL) (default) LINE (FIX)	LINE + PHONES (VOL): LINE and PHONES follow the volume LINE (FIX): LINE at a fixed full level, PHONES off
S/PDIF OUT	Off On (default)	Optical output on / off

CONFIG > DISPLAY

Item	Values	Description
AUTO OFF	60 sec 180 sec (default) 300 sec Never	Turns the display off when idle, to prevent burn-in of the OLED
ROTATION	0 deg (default) 180 deg	Turns the display and the switch directions upside down
BRIGHTNESS	1 2 (default) 3	Display brightness. A lower brightness reduces burn-in of the OLED

CONFIG > DRIVE

Item	Values	Description
READ SPEED	Max 8x (default) 4x 2x	Read speed limit of the drive
WARMUP SECTORS	0 100 200 300 (default) 500 1000	Pre-read after a disc is loaded
OVERLAP READ	Off (default) On	Read method for drives that cause clicks

Change the DRIVE items only when a drive does not play smoothly with the defaults (see Troubleshooting):

- **READ SPEED**: keep the default 8x unless there is a problem. Lower it (4x or 2x) when the drive stops or restarts at the spin-up (CHECK POWER): a lower speed needs less current and makes less operating noise (spinning sound), but leaves less margin to recover from read errors. Some drives do not read steadily at 2x, and the sound drops. Max gives the most margin, with the highest current and operating noise: it is not recommended unless the drive does not play stably at any other setting.
- **WARMUP SECTORS**: after a disc is loaded, the player reads the start of the disc once, for drives that stop for a moment on their first read. If the sound drops a few seconds into the first playback after loading a disc, set it to 0; if that does not help, try a larger value (500 or 1000). A larger value makes LOADING longer.
- **OVERLAP READ**: set it to On when clicks are heard every few seconds. Every read then overlaps the previous one and is checked, so that the drive reads from the disc instead of its cache. The drive reads more, which leaves less margin; keep it Off when there are no clicks. It has no effect when the drive itself sends wrong audio data to the player. 2x may not read fast enough for the overlapped reads: when OVERLAP READ is On, set READ SPEED to 4x or 8x.

CONFIG > PLAY

Item	Values	Description
PLAY MODE	Sequential (default) Shuffle Sequential Repeat Shuffle Repeat Single Repeat	Play order (see Play mode icons)
RESUME	On (default) Off	On: after a stop, playback starts from the stop position, as long as the power is supplied (the position is not kept while the power is off). A pause longer than 600 sec ends in a stop. Off: playback always starts from the first track, and a pause lasts until it is continued or stopped.

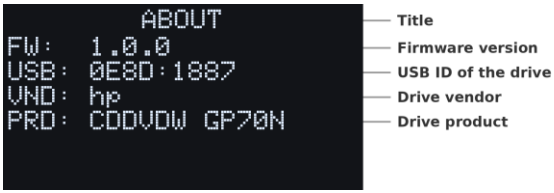
CONFIG > SYSTEM

SYSTEM holds operations on the player, not settings. Each one asks for a confirmation: NO / YES, with the cursor on NO.

Item	Values	Description
LOAD DEFAULT CONFIG	NO / YES	Restores the whole configuration and the volume to the defaults, then restarts
RESTART	NO / YES	Restarts the player

CONFIG > ABOUT

Shows the firmware version and the drive identification (USB ID, vendor, product). The USB, VND and PRD lines are shown only while a drive is connected (--- without a drive). When you contact us about a drive that does not work, please inform us of the contents of the ABOUT screen.



Example of the ABOUT screen

Troubleshooting

Symptom	Solution
NO DRIVE stays on	Check the USB cable of the drive. Connect the drive directly to CD DRIVE USB, without a USB hub. A desktop (5.25") drive needs its 12 V power supply to be connected and switched on.

Symptom	Solution
CHECK POWER, or the drive restarts when a disc is inserted	The power supply is not enough for the drive. Use a 5 V / 3 A adapter (USB power adapter PD 20W or more recommended), not a USB port of a PC. If the drive has its own power input, connect it before the USB cable. Try a lower CONFIG > DRIVE > READ SPEED (4x or 2x): a lower speed may reduce the current of the drive. If it still does not work, the drive may need more than 5 V / 3 A. Consider a power supply with a larger current capacity.
Desktop (5.25") drive: the tray does not open and no disc is detected	Connect the 12 V power supply of the drive enclosure or adapter.
LOADING stays on, or DISC ERROR	The drive cannot read this disc. Take it out and check that it is an audio CD and that it is clean. Try the disc in another drive, or another disc in this drive.
The sound drops a few seconds after the start	The drive may stop for a moment on its first read after a disc is loaded. Try CONFIG > DRIVE > WARMUP SECTORS 0 to skip the pre-read after loading. If that does not help, try a larger value (500 or 1000) so that the pre-read covers the stop.
Clicks every few seconds	The drive may return wrong data from its cache. Try CONFIG > DRIVE > OVERLAP READ On so that the drive reads from the disc and the data is checked. With OVERLAP READ On, do not set CONFIG > DRIVE > READ SPEED to 2x: it may be too slow for it.
The sound drops during playback	If CONFIG > DRIVE > READ SPEED is lower than the default, set it back to 8x: some drives do not read steadily at low speeds. If it is already 8x, the power supply may dip while the drive spins fast: try READ SPEED 4x. Check the power supply (see CHECK POWER above).
No sound from PHONES	Check the volume (VOL on the display). If LINE is shown at the bottom, set CONFIG > AUDIO > ANALOG OUT to LINE + PHONES (VOL).
UP / DOWN do not change the volume	CONFIG > AUDIO > ANALOG OUT is LINE (FIX). Set it to LINE + PHONES (VOL).
No sound from S/PDIF OUT	Set CONFIG > AUDIO > S/PDIF OUT to On. Remove the protective cap of the optical cable and select the optical input on the receiver. The receiver must accept 44.1 kHz PCM.
The display is off	Push any button to turn it on. The display turns off by itself when the player is idle and the auto off is enabled. To disable the auto off, set CONFIG > DISPLAY > AUTO OFF to Never.
The display is upside down	Set CONFIG > DISPLAY > ROTATION.

Symptom	Solution
The player does not respond	Disconnect and connect POWER again. The configuration can be reset with CONFIG > SYSTEM > LOAD DEFAULT CONFIG.
A drive or disc does not play	Not every drive works with the player (see Notes). Try another drive, or another disc.

Notes

- Only audio CDs (CD-DA) are played. Data tracks, MP3 / WAV files on a data disc, DVDs and CD-Text are not supported.
- Operation with every USB CD/DVD drive is not guaranteed. If a drive does not work, try another drive. If a disc does not play, try another disc.
- Information on tested drives is available at https://elehobica.github.io/cd_drive_player_support/
- S/PDIF OUT has no volume control. Adjust the level on the receiver.
- Connect only one CD/DVD drive to CD DRIVE USB, without a USB hub.

DISCLAIMER

The power adapter and the cables are NOT INCLUDED and must be supplied by the buyer. The CD/DVD drive is also NOT INCLUDED and must be supplied by the buyer, except with the CD/DVD drive bundle package. Operation with every CD/DVD drive, disc and power supply is not guaranteed.

Elehobica is not liable for any damage or injury to the product, the CD/DVD drive, discs, other equipment, or persons caused by incorrect connection, an inadequate power supply (including a mobile battery), or misuse of the product. This includes damage to discs, such as scratches or breakage, caused by the CD/DVD drive.

Please confirm the specifications of the CD/DVD drive and the power supply before use.

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